

CODEBOOK

Beyond the Buzzword: How do Professionals Understand and Translate Zero Trust?

THEME NAMES	CATEGORY NAMES	SUBCATEGORY NAMES	(SUB)CATEGORY DESCRIPTION	EXAMPLE OPEN CODES
ZERO TRUST SENSE-MAKING	SOURCES OF UNDERSTANDING		Mentions of reference materials or foundational resources used to understand or learn about Zero Trust principles	<i>NIST ZT documents</i> <i>CISA guidance materials</i> <i>Google BeyondCorp Model</i>
	INITIAL REACTIONS TO ZERO TRUST	—	Participants' first impressions, and emotional responses upon encountering the concept or terminology of Zero Trust	<i>makes total sense</i> <i>it sounds very high-level, and complicated</i> <i>seemed relatable and sensible</i>
	CONCEPTUAL UNDERSTANDING	ABSTRACT VIEW	Statements reflecting participants' high-level, philosophical, or principle-based interpretations of what Zero Trust means	<i>ZT-a journey; not plug-and-play</i> <i>ZT-a concept, a mindset, a philosophy; not technology</i> <i>ZT-a combination of people, process, and technology</i>
		ANALOGY	Use of metaphors, analogies, or illustrative examples to explain or make sense of Zero Trust concepts	<i>Movie theatre analogy</i> <i>It's like having car insurance</i> <i>Castle and Moat analogy</i>
		COMPARISON WITH PAST SECURITY PARADIGMS	Comparative statements that position Zero Trust against previous models	<i>ZT requires a different approach to security</i> <i>ZT is another form of defense in depth</i>
	SEMANTIC UNDERSTANDING	—	Reflections on the semantic understanding of the terminology of "Zero Trust"	<i>ZT- trust nothing is misleading</i> <i>we needed a (buzz)word with staying power</i> <i>get over the semantics of what it is called</i>
	THOUGHTS ON THE NAME - ZERO TRUST	—	Explicit opinions about the term Zero Trust	<i>misnamed, do not like it, like it / appropriate, do not care</i>
	TRUST TO ZERO TRUST	—	Reflections on how the concept of trust is reinterpreted or challenged by Zero Trust	<i>trust implies vulnerability; and is inherently flawed</i> <i>ZT - removes trust from the equation</i> <i>ZT can move the needle from trusted to trustworthy</i>
PERCEIVED BENEFITS & ADOPTION MOTIVATORS	SECURITY INCENTIVES	BETTER SECURITY PRACTICES/POSTURE	Mentions of how Zero Trust enables more proactive security practices, improving the overall security posture	<i>ZT makes security by default (by design)</i> <i>prioritizing critical assets 'crown jewels' over blanket protection</i> <i>ZT-from isolated silos to unified security</i>
		QUICKER ATTACK CONTAINMENT AND THREAT RESPONSE	Mentions of how Zero Trust facilitates faster detection, containment, and mitigation of attacks	<i>ZT-eliminates attacker lateral movement</i> <i>ZT reduces the blast radius of compromise</i> <i>ZT-protects from data breaches, ransomware, DDoS attacks</i>
		ENHANCED IAM	Mentions of how Zero Trust improves identity and access management capabilities	<i>device, identity, and resource - trust levels help determine access rights through RBAC</i>
	ORGANIZATIONAL FACTORS	BUSINESS BENEFITS	References to how Zero Trust supports broader business objectives	<i>ZT can bring least amount of interruptions in the enterprise</i> <i>the incentives from ZT are best suited for enterprises</i>
		CHANGING THREAT LANDSCAPE	Mentions of how evolving threats necessitate a shift toward Zero Trust models	<i>ZT tackles modern attacks type</i> <i>cyber attackers are very notorious (w/ huge investments in tech)</i> <i>adversaries are adept at compromising credentials</i>
		ENTERPRISE AND ASSET TYPES	Mentions of how organizational characteristics drive the relevance or urgency of Zero Trust adoption	<i>ZT crucial for hybrid workforce, increased remote users, cloud envts</i> <i>driving factors - asset value, cloud vs on-prem, access control needs</i>
	MANDATES/COMPLIANCE	—	References to external requirements that motivate or necessitate Zero Trust adoption.	<i>mandated adoption of ZT</i> <i>OMB 2209</i> <i>When they direct, we follow suit</i>
	PERCEIVING VALUE	COMMUNICATING INTENT	Mentions of how the first step involves articulating why Zero Trust matters	<i>not everyone understands ZT</i> <i>top-down approach to ensure ZT is implemented</i>
		ORGANIZATIONAL BUY-IN AND CULTURE SHIFT	Discussions around securing stakeholder commitment and fostering a cultural shift to embrace Zero Trust	<i>Demonstrate successful deployments to build credibility from people</i> <i>include UX to make security programs more effective</i>

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ZERO TRUST TRANSLATION PATTERNS	ASSESSMENT AND PLANNING	ASSESSING READINESS	Mentions of a realistic assessment of current infrastructure and capabilities as a part of the ZT Migration	ZT- readiness risk assessment-driven implementation assessment-driven tool procurement and identification
		DEVELOPING A ROADMAP	Statements around planning and strategizing the path toward Zero Trust	ZT-roadmap: assessment per pillar; per tool; per capability ZT-the FIVE pillars understand the security goals, make plan, execute in (small) steps
	PROCURING TOOLS	—	Mentions of adaptation of technical tools to enable core Zero Trust capabilities	MFA for multiple verification factors Tool-SASE;imp for ZT shift vendors customize tools based on requirements
	ESTABLISHING PROCESSES	—	Mentions of procedural changes towards ZT migration	identity and access mgnt crucial for ZT implementation having clear and correct access policies is imp moving from static authentication mechanisms to dynamic
ZERO TRUST TRANSLATION CHALLENGES	MEASURING PROGRESS	PERCEPTIONS	Comments on the Zero Trust progress measurement	ZT-a journey of constant adjustments based on evolving technology there is no ZT stamp
		MECHANISMS	Mentions of concrete methods or tools used to assess progress	CISA ZT maturity model Scorecard to measure progress
	COORDINATION ISSUES	SKILL/KNOWLEDGE/WORKFORCE CONSTRAINTS	Mentions of challenges around limited internal expertise or insufficient resources needed to support ZT implementation	upskilling and hiring rapidly is a challenge in ZT implementation skilled/robust policy admins (human-in-the-loop) are crucial
		LEADERSHIP INERTIA	References to leadership inertia hindering the momentum required for organizational Zero Trust adoption	Leadership lacks to acknowledge value of security challenge when leadership is one-directional thinkers
ZERO TRUST TRANSLATION BEST PRACTICES		CULTURAL INERTIA	Discussions around organizational and cultural inertia as a barrier to Zero Trust implementation	cultural adoption is a challenge in ZT implementation ZT can bring burden and frustration to employees not everyone understands that ZT can reduce security burden
		BUDGET AND ECONOMIC MODEL	References to financial challenges towards ZT migration	unfunded mandates budget is seen as a challenge in ZT implementation ZT is expensive to do it WELL; probably not meant for small business
	INTERPRETATION ISSUES	VAGUENESS IN REFERENCES	Statements mentioning the abstract or overly high-level nature of Zero Trust guidance materials	ZT reference materials lack practical guidance Many cybersecurity frameworks to choose from; can lead to confusion Current ZT-materials miss people part
		TRANSLATION BEYOND ENTERPRISE IT	Mentions of challenges in applying Zero Trust principles to non-enterprise or specialized environments	how can ZT concept be applied to other domains? IT / enterprise may have stricter guidelines than non-IT to implement ZT
ZERO TRUST TRANSLATION BEST PRACTICES	CONTEXTUAL ISSUES	LEGACY TOOLS AND TECHNICAL CHALLENGES	Discussions around incompatibilities with existing infrastructure, legacy systems and technical debt	ZT implementation becomes challenging with legacy systems legacy hardware and/or systems is challenging in OT and health sector
		VENDOR ISSUES	Statements mentioning vendor-related issues	Vetting vendor tools is crucial and difficult ZT is a good marketing for vendors no one stop shop for ZT
	COLLABORATION, SKILLSET, LEARNING	—	Mentions of the importance of cross-functional collaboration to support Zero Trust efforts	ZT (cybersecurity) is a team work Need for cross-functional collaboration between security and business
	STRATEGIC PLANNING AND ROLLOUT	—	Statements highlighting the value of phased implementation, and aligning ZT deployment with organizational strategy	ZT changes should be thoughtfully rolled out dont get wrapped up on ZT-terminology; stick to the basics
ZERO TRUST TRANSLATION BEST PRACTICES	BETTER POLICY GUIDANCE	—	Mentions of ZT centered on adaptive policy frameworks, and alignment with evolving risk and compliance landscapes	compliance integral to ZT, secure-by-design, and data-centric Distribute policy ownership beyond IT
	AREAS OF FURTHER RESEARCH	—	Mentions of open challenges that participants believe warrant further academic or industry investigation	more vendor-to-vendor integrations cyber psychology usability aspects of ZT